

Pareesh Madan

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TECHNICAL SKILLS

- **Languages:** Java, C++, C#, Python, JavaScript, C
- **Tools & Technologies:** Git, Docker, GCP, SQL, .NET, WPF/XAML, Entity Framework, TensorFlow, pytest

WORK EXPERIENCE

Software Developer Co-op

January 2026 – August 2026

MANTECH INC.

Guelph, ON

- Built a type-safe **User-Defined Variable** system in **C#/.NET & WPF**, driving conditional script logic
- Shipped a run scheduler and live script-monitor panel, dropping operator setup per run from **5 min to under 3**
- Integrated **4 hardware sensor families** (PeCOD, conductivity, fluorescence, pH) via Entity Framework & WPF
- Implemented runtime theme switching, replacing **300+** hardcoded colours with a centralized styling system

Undergraduate Research Assistant

September 2025 – December 2025

Arts4PD - University of Guelph

Guelph, ON

- Created an interactive **JavaScript** assessment tool that automated survey scoring across **100+ responses**
- Cut researcher manual data-entry time by **70%** by replacing spreadsheet workflows with automated scoring
- Designed a data-visualization dashboard surfacing isolation metrics in real time for the research team

Software Engineer Co-op

May 2025 – August 2025

CIBC

Toronto, ON

- Wrote **50+** automated tests in **Python** using **pytest** and **unittest**, raising test coverage by **30%**
- Added **parallel execution** via **concurrent.futures**, cutting suite runtime by **35%** in **Jenkins CI/CD** pipeline
- Integrated tests into the **CI/CD pipeline**, accelerating feedback cycles and improving deployment confidence

Software Developer Intern

May 2024 – August 2024

Bio-imaging Research Solutions Inc.

Guelph, ON

- Built a **convolutional neural network** in **TensorFlow** for disease classification at **88% validation accuracy**
- Trained on **3,500+ annotated DICOM X-ray images** of cow lymph nodes for early disease detection
- Engineered a **Python DICOM converter** automating preprocessing, cutting data prep from **hours to minutes**
- Added a **data-augmentation pipeline** (rotation, scaling, normalization), expanding the training set **4x**

PROJECTS

Bazel Open Source Contribution ❖ Java ❖ [GitHub](https://github.com)

- Contributed to **Bazel**, **Google's** open-source build system (**20k+ GitHub stars**)
- Added a **retry mechanism** with **exponential backoff** for socket failures in the repo download path
- Reduced **transient fetch failures** on flaky networks, improving build reliability
- Wrote **JUnit unit tests** covering the new retry logic and edge cases

Pokémon Team Builder ❖ Python, Flask, SQLAlchemy, REST API, Docker ❖ [Pokémon Team Builder](#)

- Engineered a **Flask** app computing net team weaknesses via set operations over a 324-cell, 18-type matrix
- Integrated the **PokeAPI REST service** with **SQL** to fetch and persist live types and sprites for **890+** Pokémon
- Built a **recommendation engine** that detects types threatening 3+ members & suggests counters for coverage

Pathfinding Visualizer ❖ Java ❖ [GitHub](https://github.com)

- Developed a desktop **pathfinding visualizer** in **Java/Swing** implementing **A***, **BFS**, and **DFS** search
- Rendered real-time grid animation with per-tile colouring of explored nodes and the final shortest path
- Added performance metrics (execution time, nodes searched) to benchmark **3 algorithms** head-to-head

EDUCATION

Bachelor of Computing, Computer Science (Co-op), Business Data Analytics Minor Expected April 2028

University of Guelph

Guelph, ON

- **Cumulative GPA:** 90% (Dean's List)
- **Relevant Courses:** OOP, Data Structures & Algorithms, Software Development, Operating Systems, Networks